MATH / BC GRADE 12

Exponential & logarithmic thinking

Connect repeated multiplication with its inverse.

Pre-calculus 12

01 Recognize exponential change

In y = a b^x, the positive base b is not 1. If a > 0, b > 1 gives growth, while 0 < b < 1 gives decay. Equal increases in x multiply y by a fixed factor.

02 Read a logarithm

log_b(A) = c means b^c = A. For real logarithms, A > 0 and the base is positive and not 1. A logarithm answers: what exponent produces this value?

03 Solve with matching bases

Rewrite both sides using the same base when possible. If b^u = b^v for a valid base b, then u = v. Check solutions against the original equation and any domain restrictions.

Worked example

Solve 2^(x+1) = 16

Rewrite 16 as 2^4. Then x + 1 = 4, giving x = 3. Check: 2^(3+1) = 16. Equivalently, log_2(16) = 4.

Watch out: log(A + B) is not generally log(A) + log(B). Logarithm rules for products do not apply to sums.

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Try it. Then check it.

Work through these questions before looking at the answers.

1. Solve 3^x = 81.

2. Evaluate log_5(125).

3. A model starts at 200 and multiplies by 0.8 each period. Find the value after two periods.

Answer key

1. 81 = 3^4, so x = 4.

2. 5^3 = 125, so log_5(125) = 3.

3. y = 200(0.8)^2 = 128. A factor of 0.8 means a 20% decrease per period.

One-minute recap

Close the notes. Explain one key idea in your own words, then retry the question you found most difficult.

About this guide

Original Study Bridge topic review prepared with AI assistance. Suggested course fit, not complete or official course coverage. Compare with your teacher's materials. Examples and practice questions are newly written.

Official course context: Pre-calculus 12

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